

Assignment 5

1. Multiple Choice Questions (Only one correct answer per question)

1. The equilibrium constant K_c for $A \rightleftharpoons B$ is 4.0. At equilibrium:
☒ a) $[B] > [A]$
b) $[B] < [A]$
c) $[B] = [A]$
d) Cannot tell
2. For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $K_c = 1.5 \times 10^{-5}$ at 298 K. This indicates:
☒ a) Strong reactant favorability
b) Strong product favorability
c) Equal amounts
d) Fast equilibrium
3. Which is omitted from equilibrium expressions?
a) Gases
b) Aqueous ions
☒ c) Pure solids and pure liquids
d) None of these
4. $Q > K$ means:
a) Reaction at equilibrium
b) Shift toward products
☒ c) Shift toward reactants
d) Impossible

5. If $K = 5.0$ for forward reaction, K for reverse is:
- ☒ a) 0.20
 - b) 5.0
 - c) 25.0
 - d) -5.0
6. Doubling coefficients in a balanced equation changes K to:
- ☒ a) K^2
 - b) $\sqrt{K}$
 - c) $2K$
 - d) No change
7. Increase in pressure by decreasing volume shifts equilibrium toward side with:
- a) More moles gas
 - ☒ b) Fewer moles gas
 - c) No change
 - d) More solids
8. Reaction is endothermic ($\Delta H > 0$). Increasing temperature will:
- ☒ a) Increase K
 - b) Decrease K
 - c) No change in K
 - d) Shift left always
9. Adding inert gas at constant volume affects equilibrium by:
- ☒ a) Shifting right
 - b) Shifting left
 - ☒ c) No shift
 - d) Increasing K
- Total P ↑
Partial P still same → K, Q same*
10. Which factor changes K ?
- a) Concentration
 - b) Pressure

- c) Temperature
- d) Catalyst

11. Catalyst affects:

- a) K value
- b) Equilibrium concentrations
- c) Rate of attainment of equilibrium *reach equilibrium faster*
- d) ΔH of reaction

12. $K_p = K_c(RT)^{\Delta n}$ for:

- a) $\Delta n = 1$
- b) $\Delta n = 0$
- c) $\Delta n = -1$
- d) All reactions

13. Strongly exothermic reaction, $\uparrow T$:

- a) Shifts to products
- b) Shifts to reactants
- c) Doubles rate constant
- d) No effect

14. Which is the Brønsted–Lowry base in the reaction $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$?

- a) NH_3
- b) H_2O
- c) NH_4^+
- d) OH^-

15. Conjugate acid of HCO_3^- is:

- a) CO_3^{2-}
- b) H_2CO_3
- c) H_3O^+
- d) OH^-

16. Which aqueous solution has the highest $[\text{H}_3\text{O}^+]$?
- ☒ a) 0.10 M HCl
 - b) 0.10 M CH_3COOH
 - c) 0.10 M NH_3
 - d) Pure water
17. At 25 °C, $K_w =$:
- a) 1.0×10^{-7}
 - ☒ b) 1.0×10^{-14}
 - c) 1.0×10^7
 - d) 1.0×10^{14}
18. pH of 0.0010 M NaOH is:
- a) 3.00
 - ☒ b) 11.00
 - c) 11.00+
 - d) 14.00
19. Weak acid HA has $K_a = 1.8 \times 10^{-5}$ at 25 °C. Its conjugate base $K_b =$?
- ☒ a) 5.6×10^{-10}
 - b) 1.8×10^{-5}
 - c) 5.6×10^4
 - d) 1.0×10^{-14}
20. Which salt gives neutral solution?
- a) NaF
 - b) NH_4Cl
 - ☒ c) NaNO_3
 - d) Na_2CO_3
21. Which is a polyprotic acid?
- ☒ a) H_2SO_4
 - b) HCl
 - c) HNO_3
 - d) HF

$$\rightarrow \frac{a}{M-a} = \frac{K_a}{a} \rightarrow \frac{M}{a} = \frac{M-a}{a} + 1 = \frac{a+K_a}{K_a} \quad \frac{a}{M} = \frac{K_a}{a+K_a}$$

22. Percent ionization of a weak acid increases when:

- a) Concentration increases
- ☒ b) Concentration decreases
- c) Volume decreases
- d) Temperature decreases

23. Which is a Lewis ^{ξV e⁻} acid?

- ☐ a) BF₃
- b) NH₃
- c) OH⁻
- ☒ d) CN⁻

24. Which binary acid is strongest?

- a) HF
- b) HCl
- c) HBr
- ☒ d) HI

25. Among oxyacids HClO, HClO₂, HClO₃, HClO₄, strongest is:

- a) HClO
- b) HClO₂
- c) HClO₃
- ☒ d) HClO₄

26. Which salt produces acidic solution?

- a) NaCl
- b) Na₂CO₃
- ☒ c) NH₄NO₃
- d) KCN

27. pH of pure water at 50 °C is:

- ☒ a) 7.0
- ☐ b) <7.0

- c) >7.0
- d) Cannot tell

28. For 0.020 M HCN ($K_a = 6.2 \times 10^{-10}$), $[H_3O^+] \approx$:

- a) 3.5×10^{-5} M
- ☒ b) 3.5×10^{-6} M
- c) 3.5×10^{-7} M
- d) 3.5×10^{-4} M

$$6.2 \times 10^{-10} = \frac{x^2}{0.02 - x} \Rightarrow x = 3.5 \times 10^{-6}$$

29. Which acid has ^{from weak acid} strongest conjugate base?

- a) HF
- ☒ b) NH_3
- ☒ c) H_2O
- d) CH_3COOH

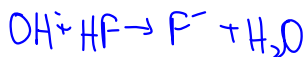
30. The pH of a buffer made from equimolar HF and NaF equals:

- a) 7.00
- ☒ b) pKa of HF
- c) pKb of NaF
- d) $14 - pK_a$ of HF

$$\frac{[base]}{[acid]} = 1$$

31. Adding NaOH to HF/NaF buffer will mostly:

- a) Precipitate HF
- ☒ b) Convert HF to F^-
- c) Increase $[H_3O^+]$
- d) Destroy buffer instantly



32. Which combination makes a buffer?

- a) HCl + NaCl
- b) $HNO_3 + NaNO_3$ } *no buffer*
- ☒ c) HF + NaF *acid - conj base*
- d) NaOH + NaCl *base + salt*

33. Buffer capacity is greatest when:
- ☒ a) $[\text{acid}] = [\text{base}]$ and both concentrations are large
 - ☐ b) $[\text{acid}] \neq [\text{base}]$ but concentrated
 - ☐ c) Dilute buffer with equal $[\text{acid}]$ and $[\text{base}]$
 - ☐ d) Only acid present
34. For titration of weak acid with strong base, pH at equivalence:
- ☐ a) < 7
 - ☐ b) $= 7$
 - ☒ c) > 7
 - ☐ d) Cannot tell
35. Ksp expression for Ag_2CrO_4 :
- ☐ a) $[\text{Ag}^+]/[\text{CrO}_4^{2-}]$
 - ☒ b) $[\text{Ag}^+]^2[\text{CrO}_4^{2-}]$
 - ☐ c) $[\text{Ag}^+][\text{CrO}_4^{2-}]^2$
 - ☐ d) $[\text{Ag}^+]^2 / [\text{CrO}_4^{2-}]$
36. If $Q > K_{\text{sp}}$:
- ☐ a) No precipitate forms
 - ☒ b) Precipitation occurs
 - ☐ c) Dissolution occurs
 - ☐ d) Solution unsaturated
37. Common-ion effect does what to solubility of sparingly soluble salts?
- ☐ a) Increases
 - ☒ b) Decreases
 - ☐ c) No change
 - ☐ d) Varies randomly
38. Which salt more soluble in acidic solution?
- ☐ a) AgCl
 - ☒ b) $\text{Pb}(\text{OH})_2$
 - ☐ c) BaSO_4
 - ☐ d) NaCl

39. Henderson-Hasselbalch equation:

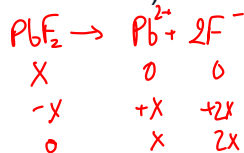
- a) $\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$
- b) $\text{pH} = \text{pK}_a - \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$
- c) $\text{pOH} = \text{pK}_b - \log\left(\frac{[\text{HB}^+]}{[\text{B}]}\right)$
- d) $\text{pOH} = \text{pK}_b + \log\left(\frac{[\text{B}]}{[\text{HB}^+]}\right)$

40. For a saturated 1:1 salt with $S = 1.0 \times 10^{-3} \text{ M}$, $K_{sp} =$:

- a) 1.0×10^{-6}
- b) 1.0×10^{-3}
- c) 1.0×10^{-9}
- d) 1.0×10^{-4}

41. Molar solubility of PbF_2 ($K_{sp} = 3.2 \times 10^{-8}$) is:

- a) 2.0×10^{-3}
- b) 1.4×10^{-3}
- c) 4.7×10^{-3}
- d) 3.0×10^{-4}



$$K_{sp} = 3.2 \times 10^{-8} = X \cdot (2X)^2 = 4X^3$$

$$8 \times 10^{-9} = X^3 \rightarrow X = 2 \times 10^{-3} \text{ M}$$

42. For acid HA, buffer 0.25 M HA and 0.15 M A^- , $\text{pK}_a = 5.20$, $\text{pH} = ?$

- a) 5.20
- b) 5.00
- c) 5.40
- d) 4.80

$$\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$$

$$= 5.20 + \log\left(\frac{0.15}{0.25}\right) = 4.98$$

2. Short-Answer/Calculation Questions

1. Predict effect on equilibrium position of: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$,

$\Delta H < 0$; when $T \uparrow$. The reaction shifts towards reactants

2. Write equilibrium constant expression for: $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$. $K_p = P_{\text{CO}_2}$

3. For $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, starting with 1.0 mol each in 2.0 L, $K_c = 50$. At equilibrium, $[\text{HI}]$ is: $\rightarrow 0.5 \text{ M}$

4. Find percent ionization of 0.010 M acetic acid, $K_a = 1.8 \times 10^{-5}$.

	$\text{H}_2\text{O}(\text{l}) + \text{CH}_3\text{COOH}(\text{aq})$	$\rightarrow$	$\text{H}_3\text{O}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
Initial	-	0.010 M	0
Change	-	-X M	+X M
Equilibrium	-	0.010-X M	X M

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

$$1.8 \times 10^{-5} = \frac{X^2}{0.01 - X}$$

$$\therefore 0.01 - X \approx 0.01 \rightarrow X^2 = 1.8 \times 10^{-7}$$

$$\therefore X = 4.24 \times 10^{-4} \text{ M}$$

$$\text{Percent ionization} = \frac{4.24 \times 10^{-4} \text{ M}}{0.01 \text{ M}} \times 100\% = 4.24\%$$

$\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$			
Initial	0.5	0.5	0.5
Change	-X	-X	+2X
Equilibrium	0.5-X	0.5-X	0.5+2X

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

$$50 = \frac{(0.5+2X)^2}{(0.5-X)(0.5-X)}$$

$$\frac{0.5+2X}{0.5-X} = 5\sqrt{2} \rightarrow X = 0.335 \text{ mol}$$

$$\therefore \text{At equilibrium, } [\text{HI}] = 0.5 + 2(0.335) = 1.17 \text{ mol}$$

5. For diprotic H_2SO_4 , 0.010 M, total $[\text{H}_3\text{O}^+]$? K_{a1} : large; $K_{a2} = 1.2 \times 10^{-2}$

1st step: All H_2SO_4 dissolves to be HSO_4^- and $\text{H}_3\text{O}^+ \rightarrow [\text{H}_3\text{O}^+] = 0.010 \text{ M}$

2nd step: $\text{HSO}_4^- + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{SO}_4^{2-}$

Initial	0.010 M	—	0	0
Change	-X M	—	+X M	+X M
Equilibrium	0.010-X M	—	X M	X M

$$K_2 = \frac{[\text{H}_3\text{O}^+][\text{SO}_4^{2-}]}{[\text{HSO}_4^-]}$$

$$1.2 \times 10^{-2} = \frac{X^2}{0.010 - X}$$

$$X^2 + (1.2 \times 10^{-2})X - (1.2 \times 10^{-4}) = 0$$

$$\therefore X = \frac{- (1.2 \times 10^{-2}) \pm \sqrt{(1.2 \times 10^{-2})^2 - 4 \cdot 1 \cdot (-1.2 \times 10^{-4})}}{2(1)} = \frac{- (1.2 \times 10^{-2}) + (2.5 \times 10^{-2})}{2}$$

$$= 6.5 \times 10^{-3} \text{ M}$$

$$[\text{H}_3\text{O}^+] = X = 6.5 \times 10^{-3} \text{ M}$$

$$\text{Hence, total } [\text{H}_3\text{O}^+] = 0.010 + 0.0065 = 0.0165 \text{ M}$$